

English Cybersecurity Glossary

Input glossary for English → Telugu translation using the IndicTrans2 Small Language Model

Context: Cybersecurity

Purpose: This glossary provides concise English term-definition pairs that can be extracted from PDF and translated into Telugu by the SLM.

1. Access Control

The process of granting or denying access to information, systems, services, or resources.

2. Authentication

The process of verifying the identity of a user, process, or device before allowing access.

3. Authorization

The process of determining whether an authenticated user or system is allowed to access a resource.

4. Availability

The condition in which authorized users can access data, information, and systems when needed.

5. Confidentiality

The property of keeping information from being disclosed to unauthorized people or systems.

6. Integrity

The property that ensures data has not been altered in an unauthorized manner.

7. Non-repudiation

The ability to provide evidence that prevents a party from denying a valid action or transaction.

8. Cybersecurity

The protection of electronic information, communication systems, and the information they contain from damage or unauthorized use.

9. Cyber Attack

An attempt to damage, disrupt, access, or compromise a computer system, network, or data.

10. Data Breach

An incident in which sensitive, protected, or confidential information is accessed, copied, disclosed, or stolen without authorization.

11. Encryption

The transformation of plaintext into ciphertext so that the original information is protected from unauthorized access.

12. Decryption

The process of converting encrypted ciphertext back into its original readable form.

13. Cryptography

The study and application of techniques for protecting information through mathematical methods.

14. Cryptographic Key

A value used by a cryptographic algorithm to encrypt, decrypt, or otherwise protect information.

15. Digital Signature

A cryptographic mechanism used to verify the authenticity and integrity of a digital message or document.

16. Hash Function

A mathematical function that converts input data into a fixed-length value called a hash.

17. Firewall

A device or program that controls network traffic to or from a system according to security rules.

18. Intrusion Detection System

A system that monitors network or system activity to detect possible unauthorized or malicious activity.

19. Intrusion Prevention System

A system that detects malicious activity and can take action to block or prevent it.

20. Virtual Private Network

A technology that creates a protected connection over a public or untrusted network.

21. Malware

Software or firmware designed or used to perform unauthorized actions that can harm a system or compromise its security.

22. Virus

Malicious software that can attach itself to other programs or files and replicate when the infected host is executed.

23. Worm

Malicious software that can self-propagate across systems or networks without requiring a host program.

24. Trojan Horse

Malicious software that appears legitimate or useful but performs unauthorized or harmful actions.

25. Ransomware

Malware that restricts access to data or systems and typically demands payment to restore access.

26. Spyware

Software designed to secretly collect information about users, systems, or activities.

27. Phishing

A social engineering technique that uses deceptive messages or websites to trick people into revealing sensitive information.

28. Spear Phishing

A targeted phishing attack designed for a specific person, organization, or group.

29. Brute Force Attack

An attack that attempts many possible passwords or keys until the correct one is found.

30. Denial of Service

An attack that attempts to make a service, system, or network unavailable to intended users.

31. Distributed Denial of Service

A denial-of-service attack launched from multiple compromised systems or devices.

32. Man-in-the-Middle Attack

An attack in which an attacker secretly intercepts or alters communication between two parties.

33. SQL Injection

An attack that inserts malicious SQL statements into an application input to manipulate a database.

34. Cross-Site Scripting

An attack in which malicious scripts are injected into web content and executed in another user's browser.

35. Vulnerability

A weakness in a system, application, process, or configuration that could be exploited by an attacker.

36. Exploit

A technique, code, or method used to take advantage of a vulnerability.

37. Security Patch

A software update designed to fix security vulnerabilities or weaknesses.

38. Zero-Day Vulnerability

A software or system vulnerability that is unknown to the party responsible for fixing it or has not yet been patched.

39. Multi-Factor Authentication

An authentication method that requires two or more different types of verification factors.

40. Password

A secret sequence of characters used to authenticate a user or protect access to a system.

41. Biometric Authentication

Authentication based on a person's physical or behavioral characteristics, such as a fingerprint or face.

42. Least Privilege

A security principle in which users and systems receive only the minimum access necessary to perform their tasks.

43. Security Policy

A set of rules and guidelines that defines how an organization protects its information and systems.

44. Incident Response

The process of detecting, analyzing, containing, and recovering from cybersecurity incidents.

45. Digital Forensics

The process of collecting, preserving, and analyzing digital evidence for investigation.

46. Security Audit

A systematic examination of security controls, configurations, policies, and practices.

47. Penetration Testing

Authorized security testing performed to identify and demonstrate exploitable weaknesses in systems or applications.

48. Personally Identifiable Information

Information that can be used alone or with other information to identify a specific person.

49. Social Engineering

The use of psychological manipulation or deception to influence people into revealing information or performing actions.

50. Threat

A circumstance or event that could potentially harm an information system, organization, or data.

51. Risk

The possibility of loss or harm resulting from a threat exploiting a vulnerability.

52. Security Incident

An event that may compromise the confidentiality, integrity, or availability of information or systems.

53. Endpoint Security

Protection of devices such as computers, smartphones, and servers that connect to a network.

54. *Cloud Security*

Policies, technologies, and controls used to protect cloud-based systems, applications, and data.

55. *Zero Trust*

A security approach that continuously verifies users, devices, and access requests rather than automatically trusting them.

56. *Security Monitoring*

The continuous observation of systems, networks, and events to identify suspicious or malicious activity.

Reference basis: Terminology was selected and phrased with reference to the NIST Cybersecurity Glossary and related NIST cybersecurity terminology resources.